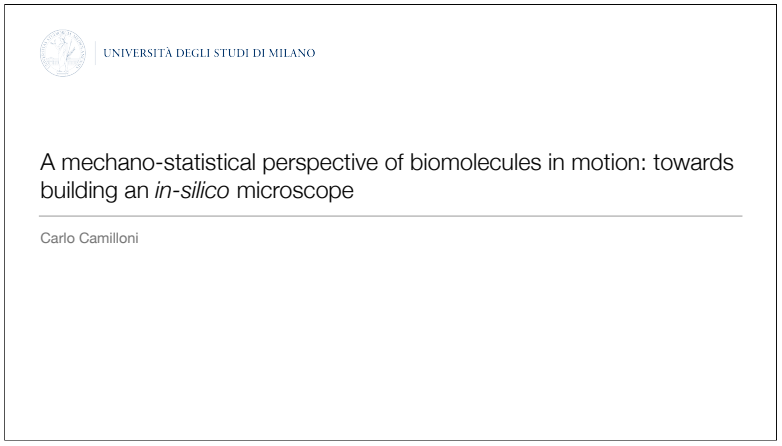
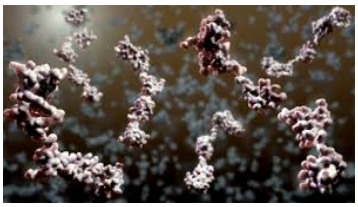


This lecture builds, step by step, a statistical-mechanical language for describing biomolecules in motion. The long-term goal is to use this language to interpret and later build the computational models used throughout the rest of the course (the ‘in-silico microscope’).

Why a Statistical and Mechanical perspective

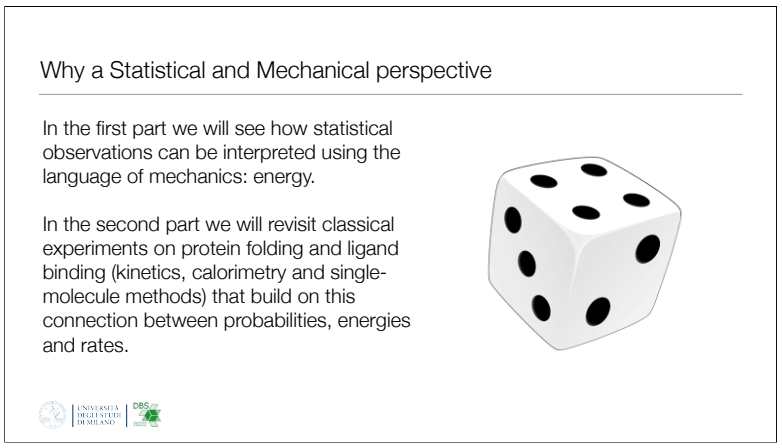
When we observe/describe biomolecules we generally handle many molecules, made of many atoms, that can:

- change their shape;
- move in 3D space;
- have interactions;



To describe all these we need to know how they move and interact (mechanics) and how to handle large numbers (statistics)

Main idea: why do we need both statistics and mechanics to describe biomolecules?
A molecular system is made of many atoms that are constantly changing shape, moving in 3D, and interacting with each other and their environment. A single, instantaneous snapshot is almost never representative on its own.
Statistics gives us tools to summarise what happens over many atoms/many observations; mechanics gives us the physical rules (forces, energy) that make some arrangements more likely than others. Combining the two lets us connect what we can observe (populations, frequencies) to what is physically happening (energies, forces).



In the first part we start from something everyone knows, rolling a die, and use it to introduce probability, sampling and averages. Then we translate the same ideas into the language of mechanics: energy, free energy and the rates of crossing barriers. The die will come back several times as our running example.

In the second part we use this framework to read classical experiments on protein folding and ligand binding. Kinetic measurements such as stopped-flow give rates. Calorimetry (DSC, ITC) gives enthalpy and entropy. Single-molecule methods (smFRET, optical tweezers) show the random hops of one molecule. Each of these measures one piece of the same picture: populations set by free-energy differences, and rates set by barriers.

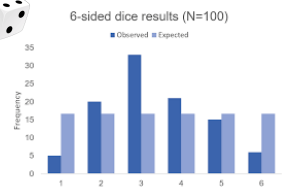
Probabilities are estimated using histograms: the problem of sampling relative and absolute probabilities



Stochastic events are estimated by SAMPLING. Sampling means exploring a probability distribution looking at many outcomes organised as histograms.

Absolute probability needs to have information on all outcomes, relative probabilities are just frequency ratios:





Absolute probability: p(i) = count(i)/(sum_k(count(k)))
Relative probability: p(j)/p(i) = count(j)/count(i)

Main idea: how do we actually estimate a probability in practice?
We never observe a probability distribution directly — we SAMPLE it, by generating many outcomes and organising them into a histogram.
Two different quantities can be computed from a histogram: the ABSOLUTE probability of outcome i, p(i) = count(i)/sum_k(count(k)), which needs the full histogram over all outcomes; and the RELATIVE probability between two outcomes, p(j)/p(i) = count(j)/count(i), which only needs their two counts.
Remember: relative probabilities require much less sampling than absolute probabilities, which is important in practice whenever full sampling is expensive.

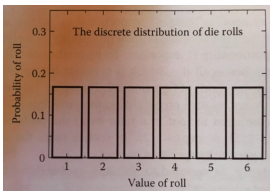
A stochastic picture for molecules in motion: probabilities



Stochastic means randomly determined, events happen by chance, by observing many events one can learn **what is the probability of the different outcomes** that is one can learn their **probability distribution**.

If some outcome is more likely than another one could think in terms of energy that that outcome is favourable in **energy**.





p(j)=1/6 with j=1 to 6 **Discrete**

$$\sum_{j=1}^{6} p(j) = 1$$

Normalised
When all possibilities are accounted for the probability is 100%

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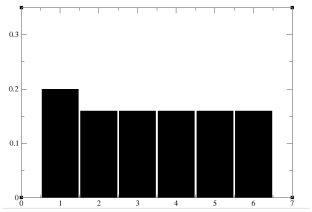
Main idea: what does ‘stochastic’ mean, and how do we represent a probability distribution, using a familiar example (a die)?
Stochastic means randomly determined: by repeating an experiment (rolling a die) many times we can estimate how likely each outcome is, i.e. its probability distribution. If one outcome is systematically favoured, we can already start to think of this in energy terms: that outcome could be said to be energetically more favourable.
A fair six-sided die gives a flat, discrete distribution, p(j) = 1/6 for every face j. It is normalised: summing over all 6 possible outcomes gives exactly 1 (100%).
Remember: any complete probability distribution must sum (or integrate) to 1 once every possible outcome is included.

A stochastic picture for molecules in motion: biases

MUST KNOW

If your die is biased towards a given number, let's say 1, then your histogram will be higher at 1. In terms of energy you can think that the state 1 is energetically more favourable.





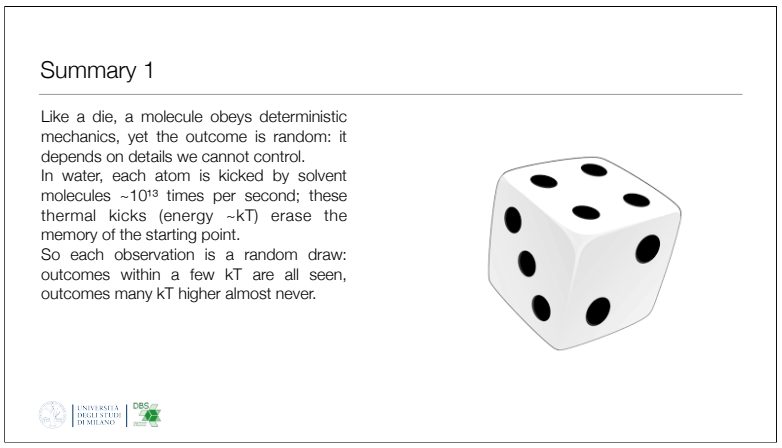
$$\sum_{j=1}^{6} p(j) = 1$$

Normalised
When all possibilities are accounted for the probability is 100%

To bias a die you change its mechanics (e.g. its centre of mass) in the presence of a force (e.g. gravity).

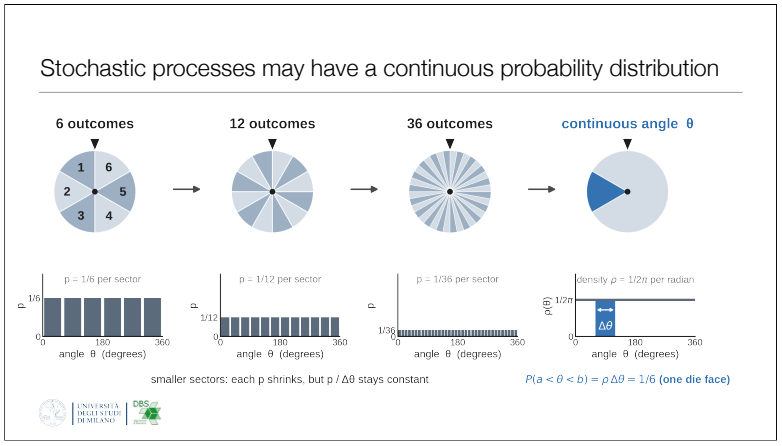
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Main idea: what happens if the die is biased, and how do we interpret that physically?
If a die favours one face (say 1), its histogram will peak there. We can describe this in energy language: state 1 is energetically more favourable than the others, the same way a ball tends to settle at the bottom of a valley.
Practically, biasing a real die means changing a MECHANICAL property of the die (e.g. shifting its centre of mass): it is the die's behaviour under gravity, not an abstract probability, that sets the outcome frequencies. This is the first concrete bridge between statistics (a skewed histogram) and mechanics (an unevenly distributed mass, acted on by gravity).
Bridge to the next slide: a die obeys deterministic mechanics, yet its outcome is random. Why? And why is the same true when we observe a molecule?

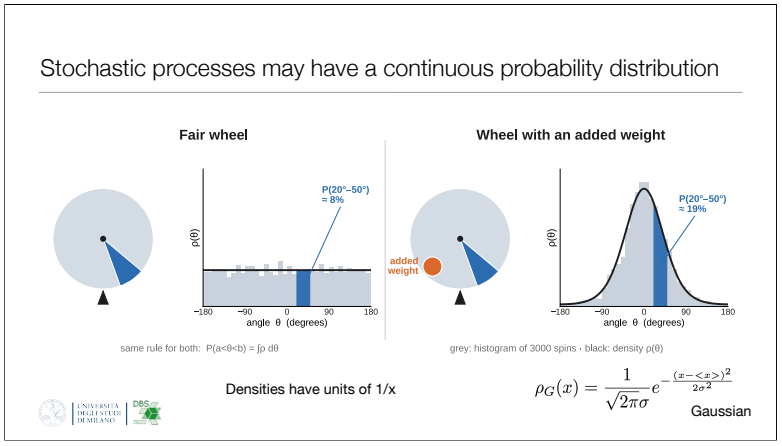


Main idea: why is the outcome of observing a molecule random (stochastic)?
A die is not random because physics is random: Newton's laws are deterministic. It is random because the outcome depends extremely sensitively on details of the throw (position, speed, spin, the surface of the table) that we neither know nor control. Two throws that look identical end on different faces.
A molecule in solution is in the same situation, only more so. It is surrounded by water molecules moving at hundreds of metres per second, and each of its atoms is kicked by them about 10^13 times per second. Each kick exchanges energy of the order of the thermal energy kT (about 2.5 kJ/mol, or 0.6 kcal/mol, at room temperature). We cannot know the positions and velocities of all these water molecules, and molecular motion is chaotic: tiny differences grow exponentially, so within picoseconds the molecule 'forgets' its precise starting point. No quantum mechanics is needed here: this is classical, thermal randomness.
The consequence: when we look at a molecule at some moment, the conformation we see is a random draw from a probability distribution. Mechanics (energy) decides how likely each outcome is.
When are outcomes really 'random'? When their energies differ by about kT or less, the thermal kicks keep exchanging between them and all of them are observed. If one outcome were many kT lower than all the others, it would be observed essentially always, and the process would look deterministic. This is why covalent bonds (hundreds of kJ/mol) do not break spontaneously at room temperature, while hydrogen bonds and other weak interactions (a few to about 20 kJ/mol) form and break all the time.
Remember: molecules behave stochastically because they are continuously kicked by a

thermal bath we do not control; energy differences compared with kT set the probabilities.



Main idea: from a discrete to a continuous probability distribution, using a wheel of fortune.
A die has 6 outcomes. A wheel divided into 6 equal sectors is the same thing. Now divide it into 12, then 36 sectors: each sector gets a smaller probability (1/6, 1/12, 1/36), and in the limit of infinitely many sectors the probability of any single exact angle goes to zero. Yet one quantity stays constant: the probability per unit angle, p/Δθ.
This is the probability DENSITY ρ(θ). For a fair wheel it is flat: ρ = 1/360 per degree (1/2π per radian). A probability is obtained only for an interval: $P(a < \theta < b) = \int \rho \, d\theta$, which is ≈ ρ·Δθ for a small interval. For example, a 60° wedge has probability 1/6, exactly like one face of the die.
Remember: for continuous variables we talk about densities, and probabilities are areas under the density curve.



Main idea: a non-uniform continuous distribution, the continuous analogue of the biased die.
Adding a weight to the wheel makes it stop more often with the weight near the bottom: the density ρ(θ) is no longer flat but peaked, here a bell-shaped curve. The histograms show 3000 simulated spins: with many spins the histogram approaches the density.
The same 20°–50° interval has probability about 8% on the fair wheel and about 19% on the loaded one: the rule P = ∫ρ dθ is unchanged, only ρ changes. In energy language, the weight makes some angles lower in energy, so they are more likely, exactly as for the biased die.
The classic bell-shaped density is the Gaussian, ρ_G(x) = exp[−(x−⟨x⟩)²/2σ²] / (√(2π) σ). Densities have units of 1/x (here 1/degree), because ρ·dx must be a pure number.
As in the discrete case, knowing a density only up to a proportionality constant is enough to compute ratios such as ρ(x1)/ρ(x2).
Link to what follows: whether discrete or continuous, an experiment usually reports only a bulk quantity, the average of an observable over the distribution, not the distribution itself. Next we see how that average is computed, what it hides (the spread), and how well we know it.

Bulk observables: Average and Standard Deviation



Experiments rarely show single outcomes, only bulk or state averages.

In practice we **estimate** averages using sums:

$$< f > \doteq \frac{1}{N} \sum_{i=1}^{N} f(x_i)$$

Variance and Standard deviation:

$$\sigma^2 = var(f) = < (f - < f >)^2 > = \int dx (f(x) - < f >) \rho(x) \doteq \frac{1}{N-1} \sum_{i=1}^{N} (f(x_i) - < f >)^2$$

Tells us about the width of the distribution $\rho(x)$.

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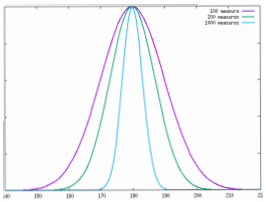
Main idea: experiments usually give us bulk quantities — averages over a huge number of molecules (or over time) — not individual outcomes.
Think of the die: we never see each molecule's roll, only the average. That average still depends on the underlying distribution: a fair die gives 3.5, a die biased towards 1 gives less, so a shift in a measured average tells us about the distribution and, for molecules, about the energy.
With a finite sample we estimate the average as a simple sum, ⟨f⟩ ≈ (1/N) Σ f(x_i). The variance (mean squared deviation from the average) and its square root, the standard deviation, tell us the width of ρ(x): how heterogeneous the individual molecules are. This is exactly what a bulk average hides.
Remember: average = what the bulk experiment reports; standard deviation = how spread out the individuals are, hidden inside that average. How well do we know the average itself? That is the next slide.

Standard Error of the Mean



Standard error: $\text{Std-err} = \sqrt{\frac{var(f)}{N}} = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1}{N^2 - N}\sum_{i=1}^{N}(f(x_i) - < f >)^2}$

Tells us about the **precision of our estimate of the average,** the idea is that our average value that we estimate from multiple observations has a resolution with the shape of a Gaussian with the width of the standard error.



Main idea: the bulk average we report is itself an estimate from a finite number of observations. How precisely do we know it? This is a different question from the previous slide's spread of individual values.
The standard error of the mean, σ/√N, tells us the PRECISION of our estimate of the average: if we repeated the whole experiment (a new set of N observations) many times, the estimated averages would themselves be scattered, following a Gaussian distribution whose width is the standard error.
Note the √N: to halve the error bar you need four times more observations.
Precision is not accuracy: the standard error only measures the statistical (random) error. A systematic error, e.g. a miscalibrated instrument or an inaccurate force field, is not reduced by collecting more data.
Remember: the standard deviation describes the individual data; the standard error describes how reliable the AVERAGE of those data is. Mixing up the two is one of the most common mistakes when reporting results.

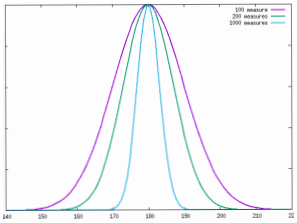
Average, Standard Deviation, Standard Error of the Mean



$$< f > \doteq \frac{1}{N}\sum_{i=1}^{N} f(x_i) \qquad \sigma^2 = var(f) = < (f - < f >)^2 > = \int dx (f(x) - < f >) \rho(x) \doteq \frac{1}{N-1}\sum_{i=1}^{N} (f(x_i) - < f >)^2$$

$$\text{Std-err} = \sqrt{\frac{var(f)}{N}} = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1}{N^2 - N}\sum_{i=1}^{N} (f(x_i) - < f >)^2}$$

The average of a random process is (for large N) distributed as a Gaussian of width std-err and so you can set confidence intervals: (±1std-err is 68%, ±2std-err 95%, ±3std-err 99.7%, …). This is the central limit theorem.





Main idea: confidence intervals, from the Central Limit Theorem.
The Central Limit Theorem states that the average of N independent observations is distributed as a Gaussian of width σ/√N (the standard error), whatever the shape of the original distribution, provided N is large enough and the distribution has a finite variance. This is why Gaussians appear everywhere in experimental science.
We can therefore use the standard error to build confidence intervals around an estimated average: about 68% confidence within ±1 standard error, 95% within ±2, 99.7% within ±3.
Two caveats. For small N the Gaussian is only approximate (Student's t distribution gives wider intervals). And the observations must be independent: consecutive frames of a molecular dynamics simulation are correlated, so the effective number of independent observations is smaller than the number of frames, and the naive error bar is too small. We will meet this again in the simulation lectures.
This is what lets us quote a bulk measurement with an error bar.

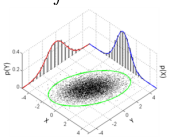
Multidimensional distribution function: projection and correlation

Probability distributions can be more than 1D. For example the probability distribution of a molecule around a receptor may be studied in terms of (x,y,z) coordinates of the center-of-mass, in this case would be 3D.

$\int \rho(x,y)dxdy = 1$ Normalisation

How can we show a multi dimensional distribution in less dimer

$\rho(x) = \int \rho(x,y)dy$ Projection/Marginalisation



$< x >= \int dxdyx\rho(x,y)$

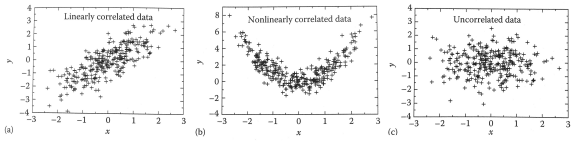
$< y >= \int dxdyy\rho(x,y)$

$< xy > \int dxdyxy\rho(x,y)$

By looking at the data can you think of a projection that is more important to understand the data?

Main idea: distributions can live in more than one dimension — how do we visualise and interpret them?
A probability distribution can depend on several variables at once: for example, the (x,y,z) position of a molecule's centre of mass around a receptor is a 3D distribution. Since we can't easily visualise more than 2–3 dimensions at once, we often PROJECT (marginalise) a multidimensional distribution onto fewer dimensions, by integrating/summing out the variables we're not interested in. The art is choosing a projection that keeps the physically or biologically relevant information while discarding the rest.

Multidimensional distribution function: correlation



$$< x >= \int dxdyx\rho(x,y)$$

$$< y >= \int dxdyy\rho(x,y)$$

$$< xy > \int dxdyxy\rho(x,y)$$

If two variables are independent then

$$\rho(x,y) = \rho(x)\rho(y)$$ **So**

$$< xy >=< x >< y >$$

So we have correlation if

$$< xy > - < x >< y >\neq 0$$

Pearson correlation coefficient: $$\frac{< xy > - < x >< y >}{\sigma_x \sigma_y}$$

Main idea: when are two variables correlated, and how do we quantify it?
If two variables are independent, their joint probability is the product of the individual (marginal) probabilities, ρ(x,y) = ρ(x)ρ(y), and then ⟨xy⟩ = ⟨x⟩⟨y⟩.
If ⟨xy⟩ – ⟨x⟩⟨y⟩ (the covariance) is not zero, the variables are correlated. Dividing the covariance by σxσy gives the Pearson correlation coefficient r, a normalised number between –1 and +1 that tells how strongly, and in which direction, two variables vary together linearly.
Careful: Pearson's r detects only LINEAR correlation. In the middle panel (non-linearly correlated data) y clearly depends on x, yet r is close to zero. Independence implies r = 0, but r = 0 does not imply independence. Always look at the data, not just at r.

Statistical Mechanics: the Boltzmann distribution



What is the connection between statistics and mechanics?

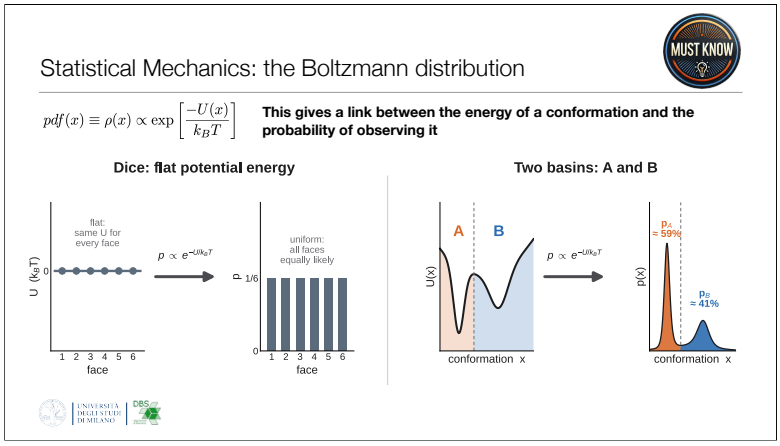
That is the Boltzmann equation: $\rho(x,v) \propto \exp\left[\frac{-E(x,v)}{k_B T}\right] = \exp\left[\frac{-U(x)}{k_B T}\right] \exp\left[\frac{-K(v)}{k_B T}\right]$

is the equilibrium distribution of a system in thermal contact with a bath at temperature T

$$pdf(v) \equiv \rho(v) \propto \exp\left[\frac{-K(v)}{k_B T}\right] = \exp\left[\frac{-\frac{1}{2}mv^2}{k_B T}\right]$$

$$pdf(x) \equiv \rho(x) \propto \exp\left[\frac{-U(x)}{k_B T}\right]$$ **This gives a link between the energy of a conformation and the probability of observing it**

Main idea: the Boltzmann distribution, the precise link between mechanics (energy) and statistics (probability), and the central concept of the lecture.
For a system in thermal contact with a bath at temperature T (e.g. a protein in water at 37 °C), the probability density of a state with positions x and velocities v is proportional to exp[–E(x,v)/kBT], where E = U(x) + K(v) is the total energy, potential plus kinetic.
Because the energy is a sum, the distribution factorises into two independent parts. The velocity part, exp(–mv²/2kBT), is the Maxwell–Boltzmann distribution: it is the same for every conformation, so it cancels in any ratio of probabilities of two conformations. The positional part, exp[–U(x)/kBT], is what matters for structure.
kBT (about 2.5 kJ/mol at room temperature) is the energy scale: a conformation 1 kBT higher in energy is e ≈ 2.7 times less likely; one 5.7 kJ/mol higher is 10 times less likely.
Up to now we only said qualitatively that low-energy states are more probable; Boltzmann makes this quantitative. Every later example in this lecture (ligand binding, protein folding) is an application of this one equation.



Main idea: two examples of how a potential energy becomes a probability distribution.
Left, the die: every face has the same energy (a flat potential), so exp(−U/kT) is the same for all faces and the distribution is uniform, p = 1/6. A flat energy means equal probabilities.
Right, a potential with two basins A and B: the probability density ρ(x) ∝ exp[−U(x)/kT] is large where the energy is low and exponentially small on the barrier. The deep, narrow basin A and the shallower but wider basin B end up with comparable populations (here about 59% and 41%): the population of a basin depends on both its depth and its width. This anticipates the free energy and the role of entropy.
Remember: low energy means high probability, the conversion is exponential, and kT sets the scale.

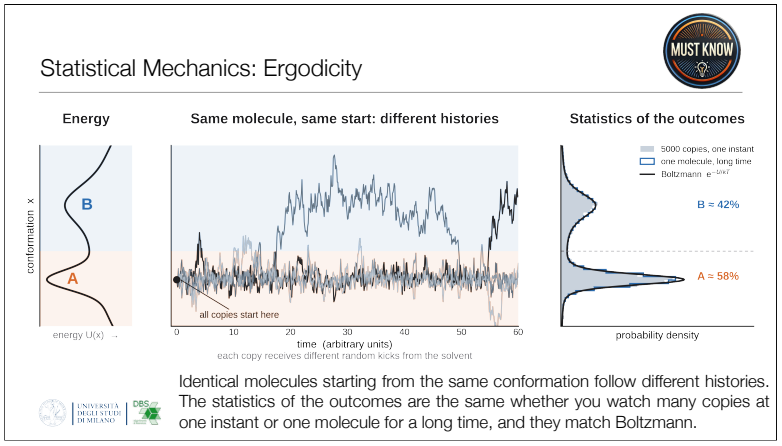
Statistical Mechanics: Ergodicity



1. **A single molecule (at constant temperature) will, given enough time, visit all the accessible conformations at random due to the collisions with the solvent. Each conformation is populated according to its energy and the temperature (the interaction with environment).**
2. **If at a given time one stops the movie of many many copies of the same system in the same condition the picture taken will display conformations according to the same statistics.**

The collection of conformations in both cases (that is the sampling) is usually called an ensemble of conformations. This ensemble is characterised first of all by the conditions in which it has been acquired (constant temperature, pressure or volume, constant number of particles, etc.)

Main idea: ergodicity, i.e. why a single molecule observed over time can tell us about a whole population.
A single molecule at constant temperature, kicked at random by the solvent, visits over time all accessible conformations, spending more time in the low-energy ones, as prescribed by Boltzmann. If instead you took a snapshot of many independent copies of the same system under the same conditions, you would see the same statistics. This equivalence between 'one system observed for a long time' (time average) and 'many systems observed at one instant' (ensemble average) is the ergodic hypothesis.
Caveat: ergodicity holds only on timescales longer than the slowest transition between states. On shorter times a molecule may stay trapped in one basin (think of a stable protein that unfolds once per hour, or a simulation too short to cross a barrier), and its time average then reflects only that basin. This is the sampling problem, which we will meet again in simulations.
The collection of sampled conformations, however it was generated, is called an ensemble, and it is always characterised by the conditions under which it was acquired (temperature, pressure, volume, number of particles, etc.).

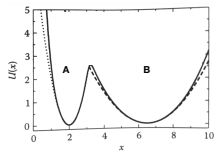


Here we simulate the same molecule many times, always starting from the bottom of basin A. Each copy receives different random kicks from the solvent, so the histories diverge immediately and some copies hop to B. If we take a snapshot of 5000 copies at one instant, the histogram of conformations (grey) matches the Boltzmann distribution e^(−U/kT) (black), with A ≈ 58% and B ≈ 42%. Following a single molecule for a long time (blue outline) gives the same histogram. This is ergodicity, and it is also why an observation is a random draw: we cannot predict which conformation a molecule will be in, only the probabilities.

Free Energy: States and Probabilities

State: is a collection of configurations (microstate), ideally b

A simple 1D potential energy



Here we can visually define two states A and B for the two basins, e.g. all the conformations belonging to A and to B

$$p_A = \int_{V_A} \rho(x)dx \propto \int_{V_A} \exp[-U(x)/k_BT]dx$$

$$p_B = \int_{V_B} \rho(x)dx \propto \int_{V_B} \exp[-U(x)/k_BT]dx$$

The Free-Energy is the "effective" energy of a state, i.e. the energy that will give the same probability

$$\frac{p_A}{p_B} = \frac{\int_{V_A} \exp[-U(x)/k_BT]dx}{\int_{V_B} \exp[-U(x)/k_BT]dx} \equiv \frac{\exp(-F_A/k_BT)}{\exp(-F_B/k_BT)}$$

$$F_i = -k_BTln\left(\int_V \exp\left[-\frac{U(x)}{k_BT}\right]dx\right) + c$$

Main idea: grouping many microscopic conformations into macroscopic STATES, and defining their free energy.
Rather than tracking every single conformation, we group together all conformations belonging to the same energy basin (basin A, basin B) into a single STATE. The probability of a state is the integral of the Boltzmann density over its region: p_A = ∫_A ρ(x)dx.
Since a state contains many conformations, its effective energy is not the energy of one configuration: it is the FREE ENERGY, defined as the energy that, if every conformation in the state had it, would reproduce the same probability: F_A = –kT ln∫_A exp[–U(x)/kT]dx.
The free energy is defined up to an additive constant (the normalisation), so only DIFFERENCES matter: F_A – F_B = –kT ln(p_A/p_B). At room temperature a free-energy difference of 5.7 kJ/mol corresponds to a 10:1 population ratio.
A wide basin contains many conformations and gets a lower free energy than its minimum energy alone would suggest: this is the contribution of entropy, which we make explicit with the biased die.

Summary: relative probabilities, energy and time

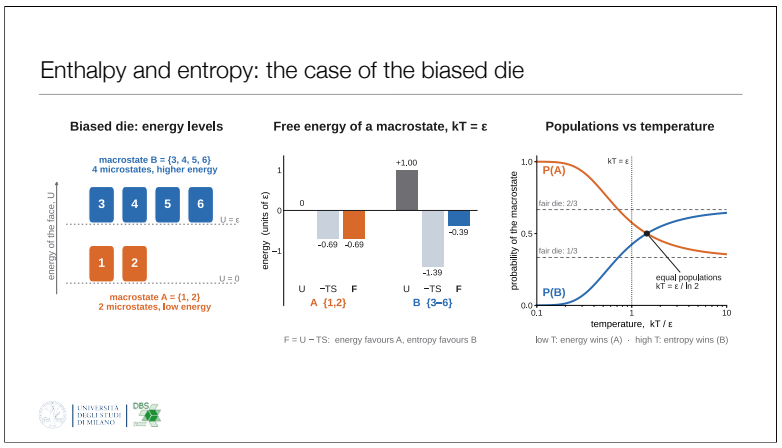


Microscopic processes are guided by **energy**, between two microscopic states their relative probability is associated exponentially to their energy difference.

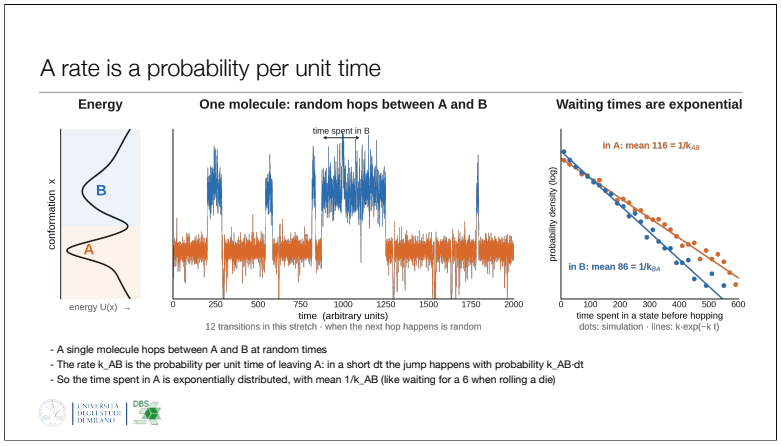
Macroscopic processes (made of many microscopic ones) are guided by **free-energy**, between two macroscopic states their relative probability is associated exponentially to their free-energy difference, the (logarithm of the) number of microscopic states in a macroscopic state is its entropy, $S = k \ln W$.

Macroscopic Processes happen in a given average time that is associated with the free-energy barrier separating them

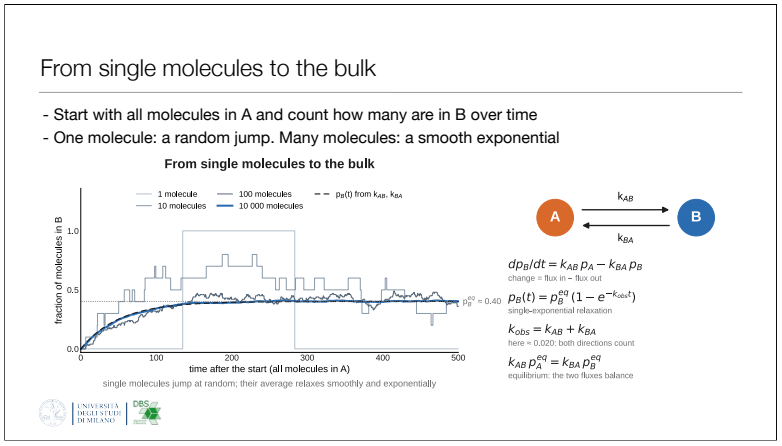
Main idea: summary, connecting probability, energy/free energy and time, at two levels of description.
At the MICROSCOPIC level (individual conformations), the relative probability of two conformations depends exponentially on their energy difference.
At the MACROSCOPIC level (states made of many conformations), the relative probability of two states depends exponentially on their FREE-energy difference. The free energy contains both the energy and the entropy of the state: $F = U - TS$, with $S = k \ln W$, where W is the number of microscopic conformations in the state.
Finally, how long it takes to exchange between states is set by the FREE-ENERGY BARRIER separating them, not by the free-energy difference between the states. This is where TIME and RATES enter the picture, as something distinct from populations: two systems can have identical populations and very different rates.



Main idea: what are enthalpy and entropy? A biased die with two macrostates.
We give each face an energy (the bias): faces 1 and 2 have U = 0, faces 3–6 have U = ε. On a real die this could be the potential energy of the centre of mass when that face lies down. By Boltzmann, each face has probability $\propto \exp(-U/kT)$.
Now group the faces into two macrostates: A = {1,2}, two microstates of low energy, and B = {3,4,5,6}, four microstates of higher energy. Energy favours A; the number of microstates favours B. The free energy of each macrostate, $F = U - TS$ with $S = k \ln(\text{number of microstates})$, puts the two effects on the same scale. At $kT = \varepsilon$ (middle panel), $F_A = -0.69\,\varepsilon$ and $F_B = -0.39\,\varepsilon$, so A wins narrowly: $p_A \approx 58\%$, $p_B \approx 42\%$.
The right panel shows how this balance depends on temperature, because the entropy term is multiplied by T. At low T energy wins and the die almost always shows 1 or 2. At high T the energy difference becomes irrelevant and the populations approach the fair-die values 1/3 and 2/3, which are set purely by counting microstates. The two macrostates are equally populated at $kT = \varepsilon/\ln 2$.
In molecules, U plays the role of the enthalpy (for liquids at constant pressure $H \approx U$). Protein folding is exactly this competition: the folded state has favourable interactions (low enthalpy) but few conformations, the unfolded state has many conformations (high entropy). This is why proteins unfold when heated.
Remember: free energy = energy − temperature × entropy; entropy counts microstates, and its weight grows with temperature.



Main idea: what a rate constant means for a single molecule.
The trajectory shows one molecule in a two-basin landscape. It rattles inside a basin for a while, then suddenly crosses the barrier. The crossing itself is fast, but *when* it happens is random. This is the same thermal randomness as before: the molecule must collect, by chance, enough energy from the solvent kicks to get over the barrier.
A rate constant k_AB is the probability per unit time of the jump: in a short time interval dt, a molecule in A jumps with probability k_AB·dt, regardless of how long it has already been in A (it has no memory). This is exactly like waiting for a 6 when rolling a die: each roll has the same 1/6 chance, so the number of rolls you wait is random, and on average it is 6. The continuous-time version of this waiting is the exponential distribution, p(t) = k·e^(–kt), with mean 1/k.
The right panel confirms it. The times spent in A and in B (dots, thousands of dwells) fall on straight lines in a log plot, i.e. they are exponential. Their means are 1/k_AB and 1/k_BA. The ratio of the mean dwell times equals the ratio of the populations.
This is what single-molecule experiments (smFRET, optical tweezers) see directly: a signal that jumps between levels at random times. In a bulk experiment we see only the average of many such molecules. That is the next slide.



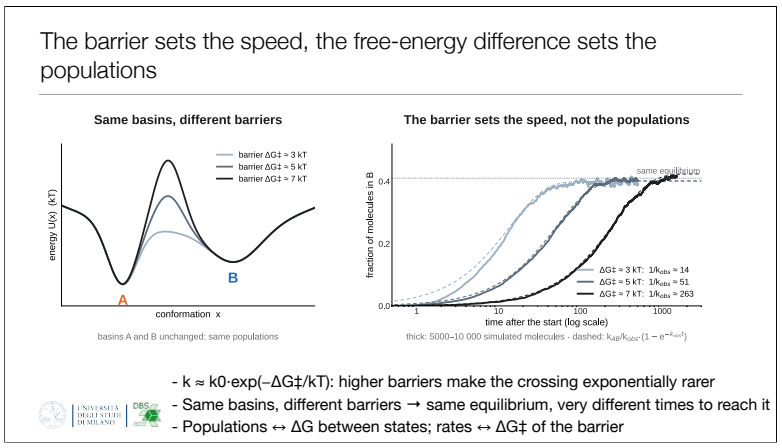
Main idea: how the random hops of single molecules become the smooth, deterministic kinetics measured in bulk.

We start at t = 0 with every molecule in A, as in a stopped-flow jump, and follow the fraction in B. One molecule gives a step: it is either in A or in B. With 10 molecules the curve is still noisy. With 100 it is recognisable, and with 10 000 it is a smooth exponential. This is the same law of large numbers we saw for the standard error: the noise decreases as 1/√N, and a real sample contains ~10^15 molecules.

The smooth curve obeys a simple rate equation: the population of B grows because of the flux from A (k_AB·p_A) and decreases because of the flux back (k_BA·p_B). Its solution is a single exponential, p_B(t) = p_B^eq·(1 − e^(−k_obs·t)), with k_obs = k_AB + k_BA. Note the sum: the system relaxes faster than either reaction alone would suggest, because both directions run at the same time. The dashed line is computed from the rates measured on the single-molecule trajectories, and it matches the simulated bulk.

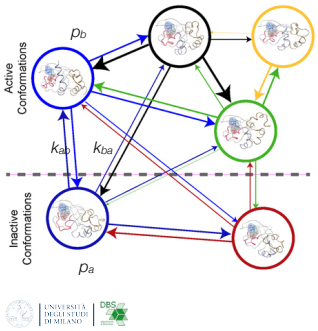
At long times the curve stops at the equilibrium value. This does not mean that the molecules stop hopping; it means that the fluxes in the two directions are equal: k_AB·p_A = k_BA·p_B (detailed balance). So p_B/p_A = k_AB/k_BA = e^(−ΔG/kT): the populations can be obtained either from thermodynamics or from the ratio of the rates.

This is exactly what we will use for folding: a stopped-flow experiment measures k_obs = k_f + k_u.



Main idea: populations and rates are controlled by different features of the free-energy landscape.
On the left we keep the two basins identical and only raise the barrier between them, from about 3 to 5 to 7 kT. On the right we repeat the bulk experiment for the three landscapes. All three curves end at the same equilibrium fraction, because the populations depend only on the free-energy difference between A and B. But the time needed to get there changes a lot: the relaxation time 1/k_obs goes from about 14 to 51 to 263 time units.
To cross the barrier, a molecule must borrow an energy ΔG‡ from the thermal kicks, and the probability of that is proportional to exp(–ΔG‡/kT). Hence k ≈ k0·exp(–ΔG‡/kT). The prefactor k0 describes how fast the molecule moves locally and depends on friction and on the shape of the barrier. In this example the prefactor also changes a little, because a higher barrier is also sharper, so the slowdown is somewhat smaller than e per kT. A useful number: at room temperature, every 5.7 kJ/mol of barrier makes a process 10 times slower.
This explains why biology can tune rates independently of stability. Enzymes lower barriers without changing the equilibrium. Two proteins can be equally stable but fold in microseconds or in seconds.

Summary: A stochastic picture of molecules in solution



Macrostate populations are associated with free energy differences:

$$p_i \propto \exp\left(-\frac{G_i}{RT}\right)$$

Exchange rates are associated with free energy barriers:

$$k_{ij} = k_0 \exp\left(-\frac{G^{\ddagger}_{ij}}{RT}\right)$$

Energy for these processes is provided by thermal energy, so essentially by collision with water molecules

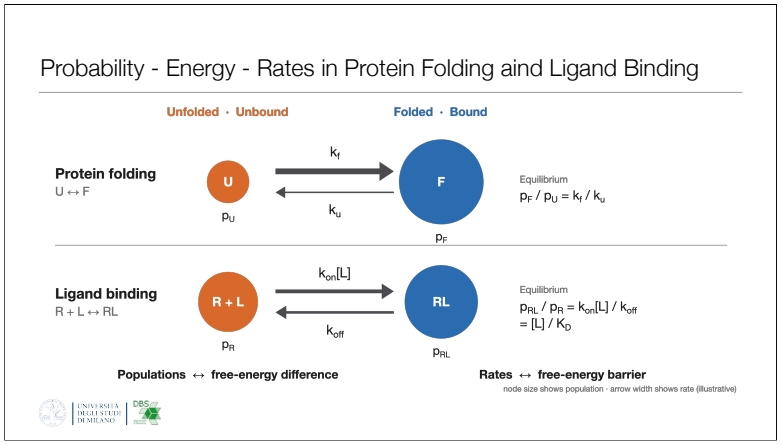
An equilibrium bulk experiment ideally captures the contribution from all the states:

$$< O >= \sum_i p_i O_i = \frac{\sum_i O_i \exp\left(\frac{-G_i}{RT}\right)}{\sum_i \exp\left(\frac{-G_i}{RT}\right)}$$

Microscopically:

$$G_i = -RT \ln \int \exp\left(-\frac{U(r)}{RT}\right) \delta(i - i(r)) dr$$

Main idea: putting it together for a molecule with several states, here a Markov state model of a protein domain.
Each macrostate i has a population p_i ∝ exp(–G_i/RT), set by its free energy. The rate of exchange from i to j is k_ij = k0 exp(–G‡_ij/RT), set by the free-energy barrier G‡_ij that separates them. The prefactor k0 depends on how fast the molecule moves locally (friction, diffusion); for protein folding it is about 10^6 s^{-1}, the folding 'speed limit' of about a microsecond.
The energy to cross barriers comes from thermal energy, i.e. from collisions with water molecules: no 'fuel' is needed, only time.
At equilibrium every exchange is balanced by its reverse (detailed balance): p_i k_ij = p_j k_ji. So rates and populations are not independent: k_ij/k_ji = p_j/p_i = exp[–(G_j – G_i)/RT].
A bulk experiment at equilibrium measures ⟨O⟩ = Σ_i p_i O_i, an average over all states weighted by their populations, while each single molecule keeps hopping between states at random.



Main idea: we now apply exactly this probability-rate-energy framework (previous several slides) to two concrete molecular examples.
Both ligand binding and protein folding are modelled as two-state (or few-state) systems: bound vs unbound for binding, folded vs unfolded for folding. In both cases we describe the system with (1) the relative POPULATION of the two states, linked to their free-energy difference, and (2) the RATES of interconversion between them, linked to the free-energy barrier.
Seeing the same mathematical structure show up in two very different biological processes is the point of this part of the lecture.

What do different experiments measure?

	Bulk thermodynamics (equilibrium spectroscopy, DSC, ITC)	Bulk kinetics (stopped-flow, T-jump, SPR)	Single molecule (smFRET, optical tweezers)
Populations, ΔG	✓	✓ via K = k_u/k_f, K_D = k_off/k_on	✓ histogram of states
ΔH, ΔS	✓ (calorimetry, or temperature dependence)	✓ from the temperature dependence of the rates	✓ from temperature or force dependence
Rates	✗	✓	✓ dwell times
Intermediates, heterogeneity, rare events	✗ averaged out	partly (extra exponentials)	✓

Every experiment measures something about the same free-energy landscape: populations are set by free-energy differences, and rates by barriers. Equilibrium (thermodynamic) experiments measure the populations, and calorimetry also measures the heats, so they give ΔG, ΔH and ΔS. However, an equilibrium average does not change in time, so it contains no information about how fast molecules exchange. Kinetic experiments perturb the system and follow how it relaxes. They give the rates, and because K is the ratio of the forward and backward rates, they give the thermodynamics too. Single-molecule experiments follow the spontaneous fluctuations of one molecule at equilibrium: the time it spends in each state gives the populations, and the time between switches gives the rates.

A stochastic picture: ligand binding

$$R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$$

If we start from a solution of free [R] and [L] we start forming RL with rate $k_{on,}$ once we have some [RL] this can also dissociate with rate k_{off}, after some time we will be at equilibrium that is that the concentration of [R], [L] and [RL] will not change anymore.

$$K_d \equiv \frac{k_{off}}{k_{on}} = \frac{[R][L]}{[RL]} = C^0 \frac{p_{unbound}}{p_{bound}}$$

The **dissociation constant** can then be defined as the ratio between k_{off} and k_{on}, it is measured in (M), where lower values means stronger binding (i.e. lower concentrations of ligand are enough to bind)

$$C^0 = 1\ M$$

$$\Delta G_{\text{binding}} = k_{\text{B}} T \ln\left(\frac{K_{\text{d}}}{C^0}\right) = k_{\text{B}} T \ln\left(\frac{k_{\text{off}}}{k_{\text{on}} C^0}\right) = k_B T \ln\left(\frac{p_{unbound}}{p_{bound}}\right)$$

This is the binding free-energy that is the amount of work that can be extracted or that must be provided for the binding reaction to happen, and you see is related to the rates as well as to the probabilities

Main idea: from rates to equilibrium, and to the binding free energy.
Starting from free R and L, RL forms at rate kon[R][L] and dissociates at rate koff[RL]. After enough time the system reaches equilibrium: binding and unbinding events still happen, but at the same rate, kon[R][L] = koff[RL], so the concentrations stop changing on average.
This gives the DISSOCIATION CONSTANT KD = koff/kon = [R][L]/[RL] (units M). Its meaning is simple: KD is the free-ligand concentration at which half of the receptors are bound. The lower KD, the stronger the binding (nanomolar KD: tight binder; millimolar: weak).
The ratio of unbound to bound receptors is [R]/[RL] = KD/[L], so it depends on how much ligand is present. To define a binding free energy we therefore choose a reference (standard) concentration C0 = 1 M: at [L] = C0, p_unbound/p_bound = KD/C0, and ΔG°_binding = kBT ln(KD/C0) per molecule (RT per mole). A KD of 1 nM corresponds to about −51 kJ/mol at 298 K.
This connects the kinetic picture (rates) back to the thermodynamic picture (free energy): the same ΔG can be read from populations or from the ratio of rates.

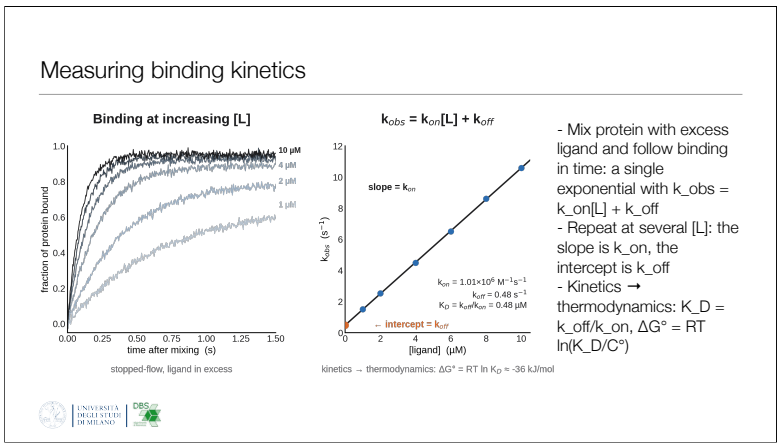
Ligand binding: a molecular example

$$R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$$

The k_{off} rate is the frequency of unbinding (s^{-1}) so that $k_{off}[RL]$ is the number of unbinding events per second per unit volume ($M\ s^{-1}$).

The k_{on} represent instead two processes (1) collision and (2) complexation. So this is the collision frequency times the complexation probability, this means that binding depends on concentration, indeed k_{on} is measured in ($s^{-1}\ M^{-1}$). $k_{on}[R][L]$ is the number of binding events per second, while $k_{on}[L]$ is the frequency of binding for a given concentration of ligand (excess of ligand).

Main idea: ligand binding as a two-state kinetic system.
We describe ligand (L) – receptor (R) binding by the probability of being unbound (R + L) vs bound (RL), together with the TIME SCALE of the process, captured by an unbinding rate (koff) and a binding rate (kon).
koff is the probability per unit time that a given complex falls apart (units s^{-1}). For a single complex, the time it stays bound is random and exponentially distributed, with mean residence time 1/koff (half-life ln2/koff). In the bulk, koff[RL] is the number of unbinding events per second per unit volume ($M\ s^{-1}$).
kon is more subtle: it combines the frequency of COLLISIONS between R and L with the PROBABILITY that a collision leads to a complex. Because collisions depend on how much ligand is around, the binding rate of one receptor is kon[L] (s^{-1}), and kon itself has units $M^{-1}\ s^{-1}$. Diffusion sets an upper limit: kon cannot exceed about 10^8–10^9 $M^{-1}\ s^{-1}$.
Example: a drug with koff = $10^{-3}\ s^{-1}$ stays bound on average for about 17 minutes. Residence time is an important property in drug design, sometimes more relevant than affinity.



Main idea: how k_on and k_off are measured, and why this also gives the affinity.
When the ligand is in large excess its concentration hardly changes, so binding behaves like a first-order reaction. The bound fraction relaxes to equilibrium with a single exponential. Its rate is the sum of the two processes, binding (k_on[L]) and unbinding (k_off), just as k_obs = k_AB + k_BA on the "From single molecules to the bulk" slide. Plotting k_obs against [L] gives a straight line: the slope is k_on and the intercept is k_off. In the simulated example we recover k_on = 1.0×10^6 $M^{-1}s^{-1}$ and k_off ≈ 0.5 s^{-1}, so K_D ≈ 0.5 µM and ΔG° ≈ –36 kJ/mol. These are the same values that ITC gives for this system: kinetics alone has given us the thermodynamics.
SPR, very common in drug discovery, measures the same thing on a sensor surface. During association the signal rises towards a plateau that depends on [L]/([L] + K_D). When the ligand is replaced by buffer, the complex dissociates and the signal decays with rate k_off, independent of concentration.
Practical limits: stopped-flow has a dead time of about a millisecond, so faster processes need temperature-jump methods (nanoseconds to microseconds). A very small k_off (hours) is easier to measure as the dissociation phase alone.

A stochastic picture: protein folding (same framework as ligand binding)

$$U \underset{k_u}{\overset{k_f}{\rightleftharpoons}} F$$

The k_u rate is the frequency of unfolding (s^{-1}) so that $k_u[F]$ is the number of unfolding events per second.

The k_f rate is the frequency of folding (s^{-1}) so that $k_f[U]$ is the number of folding events per second.

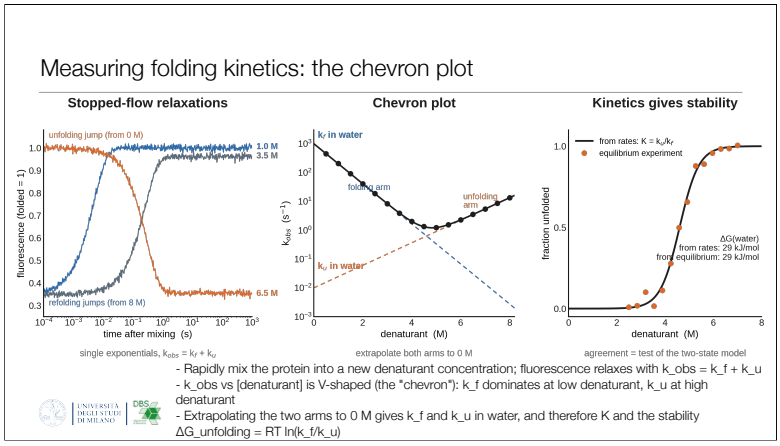
$$K = \frac{k_U}{k_F}$$

The two rates are measured by stopped-flow after a jump in denaturant concentration (next slide).

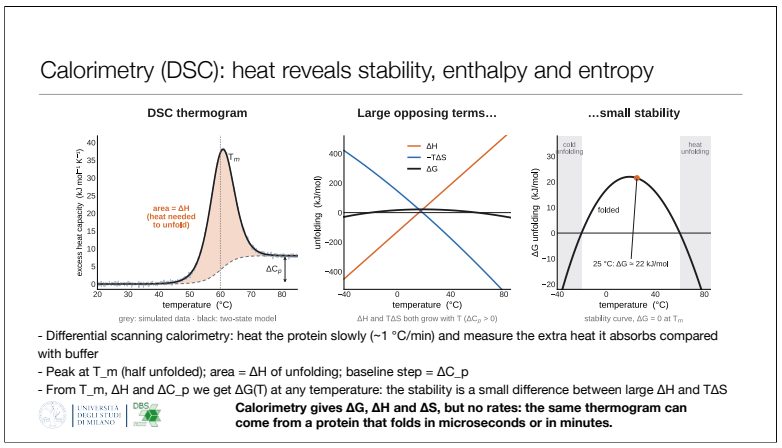
$$\Delta G_F = RT \ln K = G_F - G_U$$



Main idea: protein folding, described with the same framework as ligand binding.
Folding is a two-state process between unfolded (U) and folded (F) states, with a folding rate kf and an unfolding rate ku, both in s^{-1}: folding involves a single molecule, so no concentration appears. The number of folding events per second is kf[U], that of unfolding events ku[F].
At equilibrium kf[U] = ku[F], so the equilibrium constant is K = ku/kf = [U]/[F] (called KD on the slide by analogy with binding), and ΔG = RT ln(ku/kf) = G_F − G_U is the folding free energy, negative for a stable protein. Small proteins are typically stable by only 20–60 kJ/mol.
Experimentally, the rates are measured by stopped-flow: the protein is rapidly mixed into a new denaturant concentration and the fluorescence (often of a tryptophan) is followed as it relaxes to the new equilibrium. For a two-state system the relaxation is a single exponential with observed rate kobs = kf + ku, whatever the direction of the jump: both reactions run at the same time.
Plotting ln kobs against the denaturant concentration gives the 'chevron plot'. At low denaturant kf dominates (left arm, kobs ≈ kf); at high denaturant ku dominates (right arm, kobs ≈ ku). Each arm is roughly linear (slopes mf and mu), so it can be extrapolated to 0 M to obtain kf and ku in water. The minimum lies close to the midpoint Cm, where kf = ku and half of the molecules are folded.



Main idea: the classic folding experiment, and how it gives both kinetics and stability.
In a refolding experiment, the unfolded protein in 8 M denaturant is diluted into low denaturant. In an unfolding experiment, the folded protein is mixed into high denaturant. Fluorescence (often from a tryptophan buried in the folded state) is followed as the population relaxes to the new equilibrium. Both reactions always happen at the same time, so the observed rate is their sum, k_obs = k_f + k_u.
Denaturant slows folding and speeds up unfolding, each exponentially. On a log scale, k_obs therefore follows two straight arms meeting in a V: at low denaturant k_obs ≈ k_f, at high denaturant k_obs ≈ k_u. The minimum lies near the midpoint, where k_f = k_u and half of the molecules are folded.



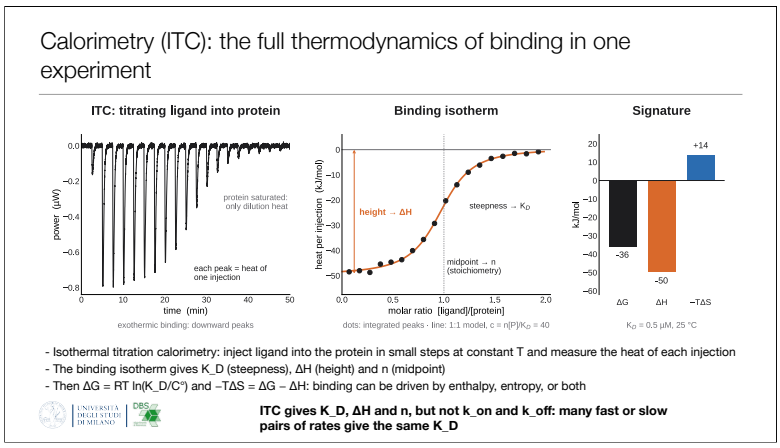
Main idea: measuring the enthalpy of folding directly, and seeing why proteins are only marginally stable.

In differential scanning calorimetry (DSC), a protein solution and a buffer reference are heated side by side, and the instrument measures how much extra heat the protein solution needs. As the protein unfolds, it absorbs heat: hydrogen bonds and other favourable contacts are broken, so unfolding is endothermic. The result is a peak in the excess heat capacity.

Reading the thermogram. The maximum is close to the melting temperature T_m, where half of the molecules are unfolded (ΔG = 0). The area under the peak is the enthalpy of unfolding, ΔH (here 350 kJ/mol, typical of a protein of about 140 residues). The baseline after the peak is higher than before: the unfolded state has a larger heat capacity, ΔC_p > 0, mainly because hydrophobic groups become exposed to water.

A check of the two-state model: the shape of the peak gives a second, 'van 't Hoff' estimate of ΔH. If it agrees with the calorimetric area, there are no significant intermediates.

Stability curve. With T_m, ΔH and ΔC_p, the Gibbs–Helmholtz equation gives ΔG at every temperature. ΔH and TΔS are each hundreds of kJ/mol and both change strongly with temperature, but they almost cancel. The stability at 25 °C is only about 20 kJ/mol, a few hydrogen bonds' worth. This is the same competition as in the biased-die slide, now with real numbers. Because ΔC_p > 0, the stability curve is a parabola: it has a maximum and falls again at low temperature. Proteins also unfold in the cold, though usually below 0 °C (cold denaturation).

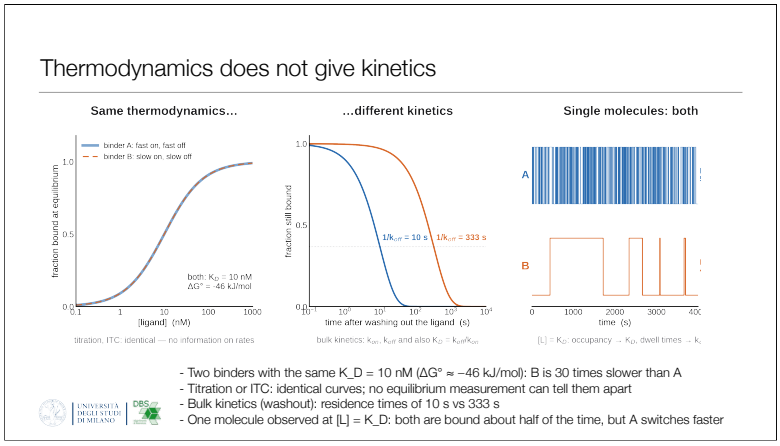


Main idea: one experiment gives affinity, enthalpy and stoichiometry, and therefore the entropy.

In isothermal titration calorimetry (ITC), the protein is in the cell (here 20 µM in 200 µL) and the ligand is in a syringe (here 200 µM). The ligand is injected in about 20 small steps, and the instrument measures the power needed to keep the cell at constant temperature. Each injection produces a peak whose area is the heat released or absorbed. At the start almost every injected ligand binds, so the peaks are large. As the protein saturates, fewer ligands bind, and at the end only the small heat of dilution remains.

Integrating the peaks and plotting the heat per mole of injected ligand against the molar ratio gives a sigmoidal binding isotherm. A fit with a 1:1 model gives three numbers. The height of the step is ΔH. The position of the midpoint is the stoichiometry n. The steepness gives K_D: a tighter binder gives a sharper step. The shape is well defined when c = n[P]/K_D is roughly between 10 and a few hundred. Here c = 40.

From K_D we get ΔG = RT ln(K_D/C°), and then −TΔS = ΔG − ΔH. In the example (K_D = 0.5 µM), ΔG = −36 kJ/mol, ΔH = −50 kJ/mol and −TΔS = +14 kJ/mol: binding is driven by enthalpy (new contacts and hydrogen bonds) and opposed by entropy (ligand and side chains lose freedom). Other complexes, typically dominated by hydrophobic contacts, are entropy-driven, because ordered water is released. Two ligands with the same K_D can have very different signatures, which is useful information in drug design. Large changes in ΔH are often compensated by opposite changes in TΔS (enthalpy–entropy compensation), just as in the DSC slide.

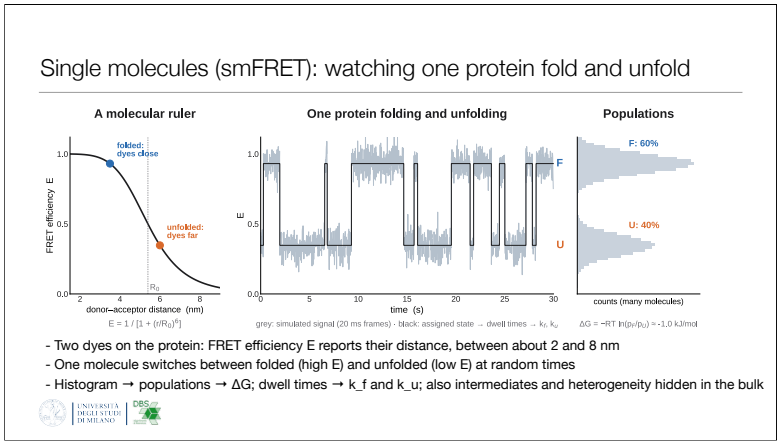


Main idea: populations depend only on the free-energy difference, while rates depend on the barrier. So equilibrium measurements cannot give rates.

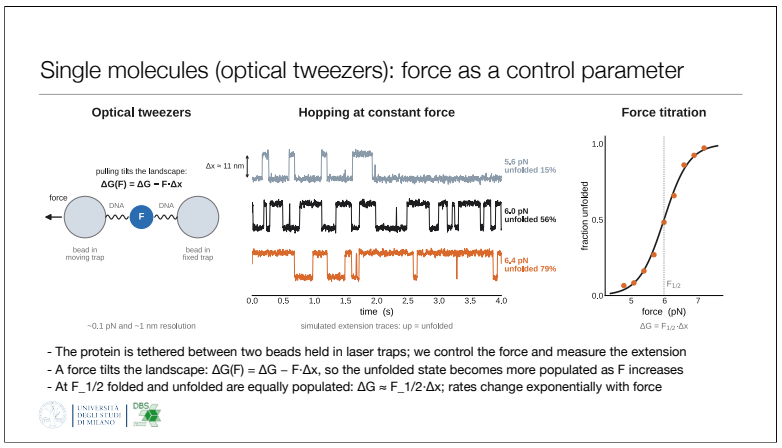
This is the binding version of the "barrier sets the speed" slide. Binder A has k_on = 10^7 $M^{-1}s^{-1}$ and k_off = 0.1 s^{-1}. Binder B has k_on = 3×10^5 $M^{-1}s^{-1}$ and k_off = 0.003 s^{-1}. The ratio k_off/k_on is the same, so K_D, ΔG and the whole equilibrium binding curve are identical (left). Equally, ITC or any equilibrium titration gives the same answer for both.

To tell them apart we must watch something change in time. In bulk, we can wash the ligand away and follow how quickly the complexes fall apart (middle): A is gone in seconds, while B stays for minutes. This matters in pharmacology: a drug's effect can last as long as its residence time 1/k_off, not as long as its affinity would suggest.

Right: one molecule observed at [L] = K_D. Both are bound about half of the time, which is the thermodynamics (here 52% and 41%). The switching frequency is completely different, which is the kinetics. The 41% for B is further from 50% because in this observation window B switched only a few times. With few events the estimate is poor, as on the standard-error slide at the start of the lecture. Slow processes need long single-molecule observations.



Main idea: seeing directly the stochastic hops that a bulk experiment averages out.
In single-molecule FRET (smFRET), the protein carries a donor and an acceptor dye. When the donor is excited, it transfers energy to the acceptor with an efficiency E = 1/[1 + (r/R0)^6], which depends very steeply on their distance r around the Förster radius R0 (about 5–6 nm for common dye pairs). E is therefore a 'molecular ruler'. In the folded state the dyes are close (high E); in the unfolded state they are far apart (low E).
Molecules are either immobilised on a surface and imaged for seconds to minutes (as in the simulated trace, 20 ms per frame), or observed for about a millisecond as they diffuse through a laser focus. Near the denaturant midpoint, one molecule switches between the two levels at random times. This is the single-molecule trajectory of the kinetics slides, now as an experiment.
From many molecules we get the histogram of E. The two peaks give the populations, and therefore ΔG = −RT ln(p_F/p_U) (here close to zero because we are near the midpoint). The times spent in each state give the rates: the mean time in U is 1/k_f, and the mean time in F is 1/k_u. Most importantly, smFRET can reveal what the bulk hides: intermediate states, subpopulations of molecules that behave differently, and even the transition paths themselves, which last only microseconds.
The same approach works for binding: label the protein and the ligand (or two partners), and bound and unbound complexes appear as different E values, for example to watch intrinsically disordered proteins fold upon binding.



Main idea: using force instead of temperature or denaturant to shift the balance between folded and unfolded, one molecule at a time.

In optical tweezers, a protein is attached through two DNA 'handles' to two micrometre-sized beads, each held in a focused laser beam. Moving one trap pulls on the protein with a force of a few piconewtons, measured with a resolution of about 0.1 pN, while the distance between the beads is measured with nanometre resolution. When the protein unfolds, the extension suddenly increases by Δx (here 11 nm).

At constant force, the molecule hops between folded and unfolded, as in our kinetics simulations. Force does work on the molecule: unfolding is favoured by F·Δx, so the free-energy difference becomes ΔG(F) = ΔG − F·Δx. Raising the force by less than one piconewton shifts the populations from mostly folded to mostly unfolded. The fraction unfolded follows a Boltzmann sigmoid. At its midpoint F_1/2 the two states are equally populated, so ΔG ≈ F_1/2·Δx: here 6 pN × 11 nm ≈ 66 pN·nm ≈ 16 kT ≈ 40 kJ/mol. In a real analysis, the elastic energy of the stretched handles and of the unfolded chain must be subtracted.

The rates also depend exponentially on force: unfolding speeds up and folding slows down as F increases, because force also tilts the barrier. From the force dependence of the rates we learn the distance to the transition state, and with enough data the whole energy landscape can be reconstructed. AFM pulling (e.g. the sawtooth unfolding of titin domains) and magnetic tweezers apply the same idea to larger forces or longer times. Ligand binding can be probed in the same way, by measuring the force needed to break a complex, or how a bound ligand stabilises the folded state.

Recap

- Observing a molecule is a random draw: thermal kicks make molecular outcomes stochastic
- Microscopic probability ↔ energy: $p \propto \exp(-E/kT)$
- Macroscopic (free-energy) probability ↔ entropy and energy of a state
- A rate is a probability per unit time: single molecules hop at random, the bulk relaxes exponentially
- Populations ↔ free-energy differences; rates ↔ free-energy barriers
- Ligand binding and protein folding: same framework, two different molecular events
- Bulk kinetics gives the rates and, through K = k_forward/k_backward, also the thermodynamics
- Bulk thermodynamics (equilibrium curves, calorimetry) gives ΔG, ΔH and ΔS, but no rates
- Single-molecule experiments give both, plus what averages hide



Main idea: recap of the lecture's logical chain.
(1) Observing a molecule is a random draw: thermal kicks from the solvent make molecular outcomes stochastic, and only their probabilities can be predicted.
(2) At the microscopic level, probability and energy are linked exponentially by the Boltzmann distribution.
(3) At the macroscopic (state) level, probability is linked to the FREE energy, which bundles together the energy and the entropy (number of microstates) of a state.
(4) A rate is a probability per unit time: a single molecule hops at random, with exponentially distributed waiting times, while the bulk relaxes smoothly and exponentially, with kobs equal to the sum of the forward and backward rates.
(5) Rates are governed by the free-energy BARRIER, populations by the free-energy DIFFERENCE.
(6) Ligand binding and protein folding are two concrete examples of the same framework.

The common theme of Structural Bioinformatics: sampling and energy (or score, or loss function)



In the next lectures we will see how the problem of SAMPLING and the problem of the ENERGY or SCORING FUNCTION or LOSS FUNCTION, are interconnected and always present. These are the problems to cope with when doing MD simulation, structure prediction, docking, protein design, …

You can think of the next lectures as specific strategies to cope with these two problems when trying to address specific problems. All the algorithms we will see are related to them (MD, MC, EM, ML/AI, …)

Main idea: bridging to the rest of the course — sampling and energy/scoring/loss are the two recurring problems behind every method you will see.
Every computational method in structural bioinformatics — molecular dynamics, Monte Carlo, structure prediction, docking, protein design, machine learning — ultimately has to solve two intertwined problems: how to SAMPLE the relevant states efficiently, and how to define/evaluate an ENERGY (or score, or loss function) that tells you which states are good.
You can think of the rest of the course as a tour of different strategies for tackling these same two problems in different contexts.

Questions:

- How do you calculate the average outcome of a stochastic process?
- What is the difference between the standard deviation and the standard error of the mean?
- Which is the probability distribution function of the average of a process?
- What is the difference between the absolute probability of an event and the relative probability of that event with respect to another?
- What is the relationship between the probability of a macrostate and its free-energy?
- What is the relationship between the probability of a configuration and its energy?
- What is the free energy change of a process?
- …



Further questions on stochasticity and kinetics:
Why is the outcome of observing a single molecule stochastic?
What does a rate constant mean for a single molecule, and how is it related to the average time spent in a state?
Why do the populations of a two-state system depend on the free-energy difference, while the rates depend on the barrier?
In a stopped-flow experiment on a two-state protein, how is the observed rate kobs related to kf and ku?
What is the meaning of KD in terms of ligand concentration?